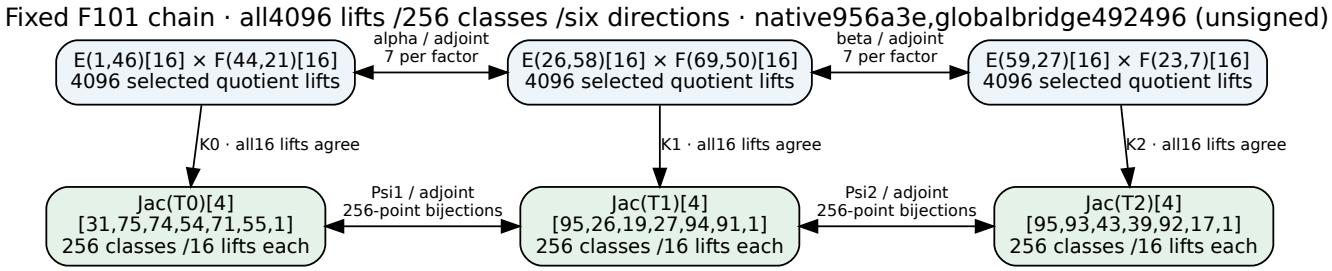
Whole 4-primary maps without pointwise division

2026-11-12 · fixed F101 chain · scientific standing unsigned

All six first,second and compressed forward/adjoint directions now have a complete quotient evaluation on every256 geometric $A_i[4]$ classes. For each model,the checker retains4,096 elliptic-product16-torsion lifts, and all16 representations per class give the same result. Full changed native32/956a3e passes with zero failures,ignores or filters;old guards, formatting,strict Clippy and all15 source/fixture hashes pass. The complete 55,041-byte raw stream is preserved as standard XZ8,252,with raw/archive digests and exact roundtrip. These are correctness checks. [pcc:unverified]



Compressed forward and adjoint:both exact two-stage maps on every256 classes

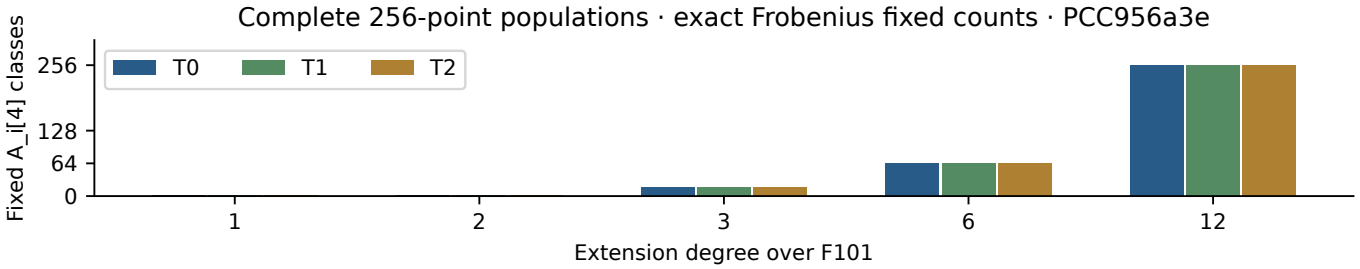
A quotient route on the difficult primary component

The admitted global identity $[4]F=K_j\varphi H_i$ and $\mathrm{norm}H_iK_i=[4]$ imply $F\,K_i=K_j\varphi$ by cancellation in Hom. This does not cancel4 on point groups. If $x\in A_i[4]$,any preimage b under K_i satisfies$[4]b\in\ker K_i\subset B_i[4]$,so $b\in B_i[16]$. The actual16-point4-graph selects all4,096 allowed lifts. Kernel transport makes $K_j\varphi(b)$ independent of the chosen lift. TWO_PRIMARY_QUOTIENT_PROOF.md/492496 gives the written global bridge. [pcc:unverified]

Complete halving fibres give64 then256 checked points in each elliptic factor. Nonzero2-torsion has repeated abscissas and both ordinates;origin has its complete four-point fibre. The old ramified quartic pullback and separate Cantor engines are reused. No new field/group engine or producer Balanced/ Richelot/trace evaluator is imported. [pcc:unverified]

Full-domain checks and the field requirement

All65,536 Cantor additions pass on each256-class population. Every direction passes all65,536 group-law pairs,norm7 or49,Frobenius equivariance and all16 actual gluing-graph transports. Both compressed maps equal the actual two stages on all256 classes. Factor16-torsion norms and256-pair Weil-similitude tables also pass. [pcc:unverified]



Frobenius acts through all elliptic lifts;coordinatewise Frobenius in the non-F101 quintic chart would be invalid. Only64 of256 $A_i[4]$ classes are fixed over degree6;all256 are fixed over degree12. The full action has exact order12,whereas each elliptic4-torsion factor needs degree6. This is a field-of-definition result for the fixed groups. [pcc:unverified]

Precise failures and a cross-family degree obstruction

The native doubling control [2]F has image16,zero fibre16 and255 differences from F in every direction. Its historical variable name is wrongquarter2. The literal old norm/pull map is insteadL=K_jφH_i=[4]F, which is zero onA_i[4]. Every integer multiplier[c]L remains zero. ProtocolV2 distinguishes these maps explicitly. The latter is proved as a written/Lean consequence of the global factorization;it is not reported as a new direct native H evaluation. Native32 is unchanged. [pcc:unverified]

Crossing C-type and T-type requires a factor4 in degree

The prior cross-Hom0 and actual factor embeddings make the two connected elliptic subvarieties intrinsic. Their canonical sum degrees are4 on Jac(C_i) and16 onJac(T_j). A separable geometric isogeny of degree n restricts to factor isogenies of degrees r,s,each dividing n. The commuting addition square gives n·sum_degree(source)=sum_degree(target)·rs. [pcc:unverified]

From C to T,this yields n=4rs. From T to C,it gives rs=4n. Write n=ru=sv;then n^2=rsuv=4nuv,and cancel positive n to obtain n=4uv. Thus every cross-family separable geometric isogeny degree is divisible by4 in either direction. The actual Richelot maps and their adjoints have degree4;composing with the odd-degree49 C-family routes attains 2-part exactly4 for every cross-index pair. The six surfaces consequently form two components under separable odd-degree geometric isogenies. Classification,embeddings,global degrees and separable kernel orders are explicit written dependencies,not deductions from the finite census. [pcc:unverified]

Six faithful formal consequences

TwoPrimaryQuotient proves the halving polynomial(ace317),quotient lift independence(807a28),16-torsion preimages(32c9f0),Frobenius^{12} modulo16 (5dc179),reverse4-degree divisibility(919e88),and all integer cleared-map multipliers vanishing on4-torsion(cf3d5a). Each is individually bound and locally passes;gate05 checks six with zero trusted native proofs. The written census is160 declarations across32 modules. Actual varieties, Picard maps,Hom cancellation and geometric kernel identification remain written bridges. [pcc:unverified]

The native identity adapter reuses unchanged repository issuer/checker/SHA code. Halving IDC1h26ac6af51218fa21 finalversion03 has32 evaluations and 64 extra domain-separated checks,with the exact checked Lean-file digest. All23 algebra tests pass;old duplication IDC1he156b6adc8f9cdd5 replays byte-identically. e13811 retains complete source/raw/hash evidence. Finite-field evaluation remains separate from the universal Lean proof. [pcc:unverified]

Sources and preserved attempts

Primary Galbraith9.1 equation9.2 and MilneI7,I8.10–12,IV2.2 were read, alongside old quotient89e13d/ actual4-glue933d35 and cross-Hom b05286. KB/wrapper and Conductor are unavailable;primary/local fallback is recorded. No novelty claim. All original31 native tests and prior figures keep their historical scopes;these figures show the new route. [pcc:unverified]

Failed formal02/ede154 and04/5d81f0 retain full source/gate/build/verdict evidence. Final proof uses standard congrArg and zsmul_zero lemmas. Certificate01/02 remain immutable;02 points to the failed module and is not an accepted formal pointer. The first bound claim is reused with its unchanged declaration digest;failed output never counts as a proof. [pcc:unverified]

Consumer1508/c712 exactCI43202427032 succeeds98reference+five/full700efe, root mirror37223f. Root92 exact43210021347 succeeds30/14930/full2e26e7. Published4e scope31+23/15431 has own43215479664,next13:01;this new32+ 23/16032 scope needs its own checkpoint and CI. Next:integrate the independent checks into a fresh consumer child,then bounded elliptic cycle/gluing closure. Live-budget/ S3 admission and atlas-vault access remain blocked;native correctness/proof work continues. [pcc:unverified]